EX. NO: 5 IMPLEMENTATION OF RSA ALGORITHM

AIM

To implement the RSA encryption and decryption algorithm using Python.

DESCRIPTION

RSA (Rivest–Shamir–Adleman) is an asymmetric cryptographic algorithm that uses a pair of keys: a public key for encryption and a private key for decryption. It provides secure communication by ensuring that only the intended receiver can decrypt the message.

ALGORITHM

STEP-1: Choose two prime numbers **p** and **q**.

STEP-2: Calculate $\mathbf{n = p \times q}$.

STEP-3: Compute $\mathbf{\varphi(n) = (p - 1) \times (q - 1)}$.

STEP-4: Choose a public key **e** such that $\mathbf{1 < e < \varphi(n)}$ and $\mathbf{gcd(e, \varphi(n)) = 1}$.

STEP-5: Compute the private key **d** such that $\mathbf{(d \times e) \bmod \varphi(n) = 1}$.

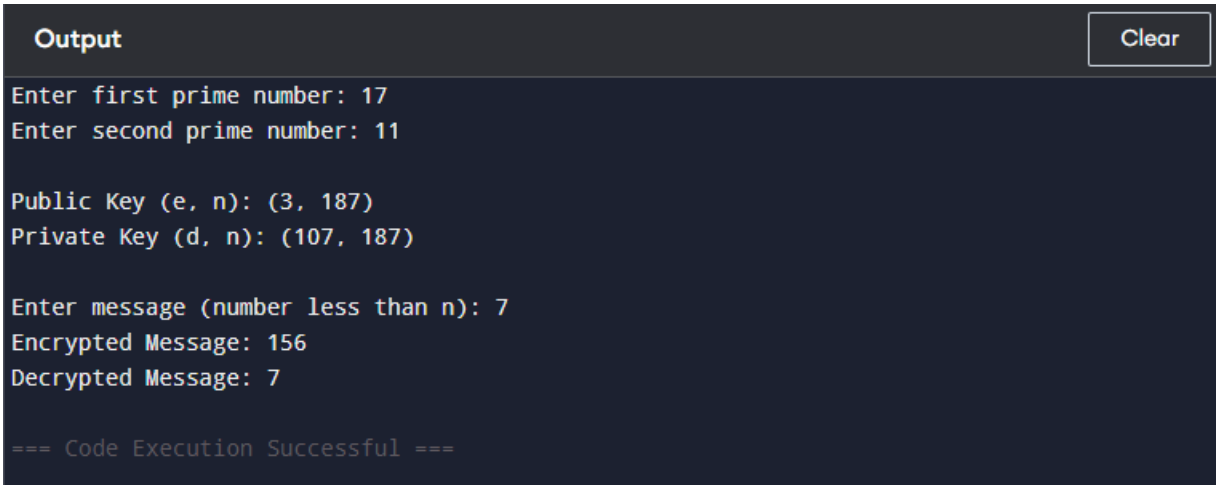
STEP-6: Encrypt the message using the public key.

STEP-7: Decrypt the ciphertext using the private key.

CODE

```
from math import gcd


# Function to find modular inverse

def mod_inverse(e, phi):

    for d in range(2, phi):

        if (d * e) % phi == 1:

            return d


# Input two prime numbers

p = int(input("Enter first prime number: "))

q = int(input("Enter second prime number: "))
```

```
n = p * q
phi = (p - 1) * (q - 1)

# Select public key
e = 2
while e < phi:
    if gcd(e, phi) == 1:
        break
    e += 1

# Generate private key
d = mod_inverse(e, phi)

print("\nPublic Key (e, n):", (e, n))
print("Private Key (d, n):", (d, n))

# Message
msg = int(input("\nEnter message (number less than n): "))

# Encryption
cipher = pow(msg, e, n)
print("Encrypted Message:", cipher)

# Decryption
plain = pow(cipher, d, n)
print("Decrypted Message:", plain)
```



RESULT

Thus, the RSA encryption and decryption algorithm was implemented successfully using Python.